Patterns in Large Language Model Reasoning Across Languages

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Abstract

Modern LLMs are almost always capable of both communication, and reasoning multi-lingually—that is, reasoning in a multitude of languages—but does the language of reasoning affect the model's performance as a whole? This paper investigates exactly the relationship between reasoning language, and performance. We test Anthropic's Claude Opus 4.8, Google's Gemini 3.1 Pro, and DeepSeek-V4-Pro—two Western and one Chinese frontier models—on 30 problems each from the HMMT (Harvard-MIT Math Tournament) and 全国高中数学联赛一试 (the fill-in-the-blank first round of the China National High School Mathematics League), constraining their input, reasoning, and output to either English, Chinese, or no language at all totalling 720 calls. Across the board, Chinese prompting and reasoning lead to the highest accuracy, with Gemini 3.1 boasting the highest per-model accuracy. DeepSeek has a preference of Chinese reasoning, while the Western models will almost always avoid the language. Vitally, the experiment implies that the function of language in a model's reasoning is more load-bearing than incidental.

1 Introduction

Many frontier Large Language Models (LLMs) of the modern AI sphere are trained overwhelmingly on English data [1], and yet are able to operate *multi-lingually*; while most information, reasoning technique, and strategy is accumulated and learned from English data, modern models will almost always respond—and continue to converse—in the language originally prompted in.

Furthermore, innovations in the field of LLMs enable a model to complex, multi-step reasoning when faced with challenge—one such architecture is called, fittingly, "chain-of-thought" [4]. This architecture sees models taking a series of steps to work through a task, and produce a highly accurate and efficient output. Often, the logic and process for this reasoning is rooted in English data and context, but is that necessarily the optimal strategy for performance? If a model's reasoning were to be constrained to another, structurally and phonetically distinct language, how might its performance differ? *What patterns emerge in the reasoning of LLMS across languages?* This paper aims to delineate the relationship between language, and performance in reasoning, through experimentation in English and Mandarin Chinese on complex, high-level mathematics problems.

2 Methods

We assessed three frontier reasoning models with four prompt conditions each, over a sixty-problem bilingual benchmark—a $3 \times 4 \times 60$ design yielding a total 720 runs. This experiment aimed to evaluate whether constraining reasoning to a single language changes accuracy, as well as to determine the languages naturally adopted by each model when left.

2.1 Models and inference

Models evaluated include Claude Opus 4.8 (`claude-opus-4-8`), DeepSeek-V4-Pro (`deepseek-v4-pro`), and Gemini 3.1 Pro (`gemini-3.1-pro-preview`) through their official APIs in early June 2026, each at its maximum exposed reasoning setting: Claude with adaptive thinking at effort `high`, DeepSeek with `reasoning_effort high` and thinking enabled, and Gemini with `thinking_level` HIGH and thoughts (reasoning trace) included. Every call used a 32,768-token output budget and was streamed; this streaming was made necessary due to server connection drops amidst long generations on certain difficult problems. Sampling temperature was *not* explicitly configured, and was left at provider defaults across the board; Opus 4.8 removes the option, DeepSeek-V4-Pro does not expose it, and reasoning-mode Gemini is notoriously unstable at zero temperature.

2.2 Benchmark

The employed benchmark is composed of sixty short-answer competition problems: thirty English-native and thirty Mandarin-native, each rendered in both languages such that the identical item may be posed in either. The English-native half is from the Harvard-MIT Mathematics Tournament (HMMT) February (fifteen problems each from 2025 and 2026, via MathArena); the Mandarin-native half comes from 全国高中数学联赛一试 (the fill-in-the-blank first round of the China National High School Mathematics League, 2024 and 2025, A and B papers). All problems require symbolic answers—namely fractions, radicals, factorials, finite sets, intervals, and disjunctions—rather than multiple-choice options, or bare integers. American Invitational Mathematics Examination (AIME) problems were excluded despite their convenience, due to the answer space compression resulting from all answers being integers $\in [0, 999]$; this restriction not only invited guessing, but were also approached at ceiling by these models. Likewise, proof-based formats—IMO-style problems and the CNHSML second round (二试) among them—were excluded, due to the invited possibility for grader variance, thus hindering the objectivity of model evaluation.

2.3 Conditions

Each problem was run under four conditions, merging language constraint with prompt language: the two *constrained* conditions explicitly instruct the model to reason and answer entirely in a single language (English-constrained, Mandarin-constrained), while the two *unconstrained* conditions present the same statement, prompted in either English or Mandarin with no language instruction at all:

> *English-constrained:* "Think step-by-step in only English. Your reasoning and final answer should both be in English. Use LaTeX throughout your solution. Box your final answer with `\boxed{}`."

> *Mandarin-constrained:* "仅用中文思考。你所有的推理、过程和最终答案都应该只用中文。在你的解答中使用 LaTeX 语法。请用 `\boxed{}` 框出你的最终答案。"

The two unconstrained prompts are these with the language constricting instructions deleted, so that the model receives only the proper answer format, and the problem itself, in both languages.

3 Results

Across all three models, constraining the reasoning language had some but no dramatic affect on performance; the substantial differences in the data lie between the three models, not between the conditions.

3.1 Accuracy by condition

Table 1 reports accuracy for every model-condition cell. Within each model, the four conditions landed within a few percentage points of one another; a spread of seven points for Claude (48-55%), five for DeepSeek (67-72%), and seven for Gemini (85-92%)

The variation in the table is the result of capability rather than language: overall accuracy ranges from Gemini at 88% through DeepSeek at 69% to Claude at 52%: a roughly thirty-six-point span, that dwarfs any within-model condition difference.

Despite the absence of any significant patterns, it is notable that the Chinese-only condition was the single highest-scoring condition for every model (Claude 55%, DeepSeek 72%, Gemini 92%). The margins are small—at most, just a few points over the next-best condition. Nevertheless, this pattern is present among all three independently trained models.

Model	EN-only	ZH-only	Unc. (EN)	Unc. (ZH)	Overall
Claude Opus 4.8	48%	55%	52%	53%	52%
DeepSeek-V4-Pro	67%	72%	68%	68%	69%
Gemini 3.1 Pro	88%	92%	85%	85%	88%

Table 1: Accuracy by model and language condition ($n = 60$ per cell).

3.2 Reasoning and answer language

Model compliance with the language instruction component of the prompt depended largely on whether we looked at the model's final answer, or the reasoning trace. Under the Chinese-only condition, all three models produced their *final answer* in Mandarin at high rates—Claude at 78%, DeepSeek at 73%, and Gemini at 98% (see Figure 1)—so obedience at the level of the answer was broad. By contrast, the reasoning trace, diverged: only DeepSeek really reasoned in Mandarin consistently (93% of traces), while Claude (3%) and Gemini (23%) returned almost exclusively English-language traces. However, as noted in the Methods, Claude and Gemini expose only a summarized trace rendered in English; for those two models, the Mandarin-reasoning figures reflect only the summarizer and cannot be read as the model's true internal language.

The same asymmetry is visible in the unconstrained results. Given an originally Chinese problem with no language instruction, DeepSeek's raw trace and final answer stayed predominantly in Mandarin, whereas the surfaced output from Claude and Gemini leaned English or mixed (see Figure 2).

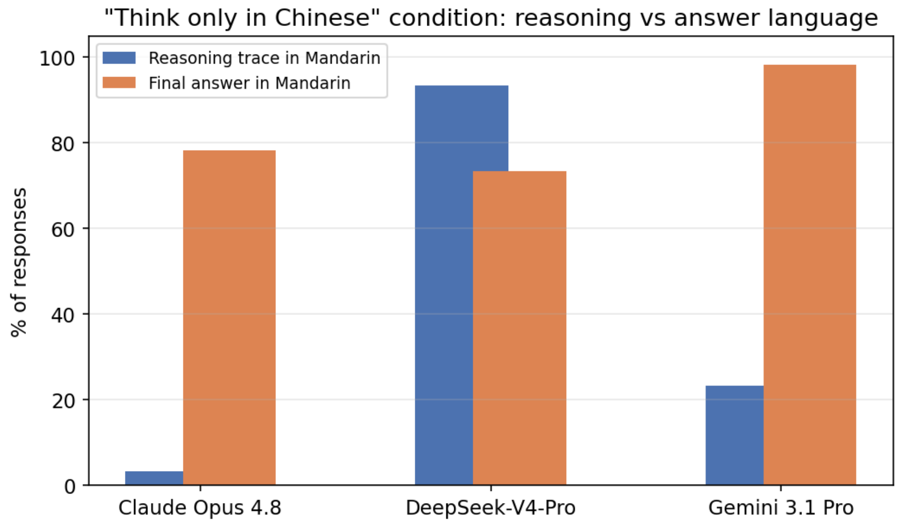

Figure 1: Under the Chinese-only condition, the share of responses whose final answer versus whose reasoning trace appears in Mandarin. All models comply in the final answer, but only DeepSeek's raw trace is in Mandarin; Claude's and Gemini's summarized traces return in English.

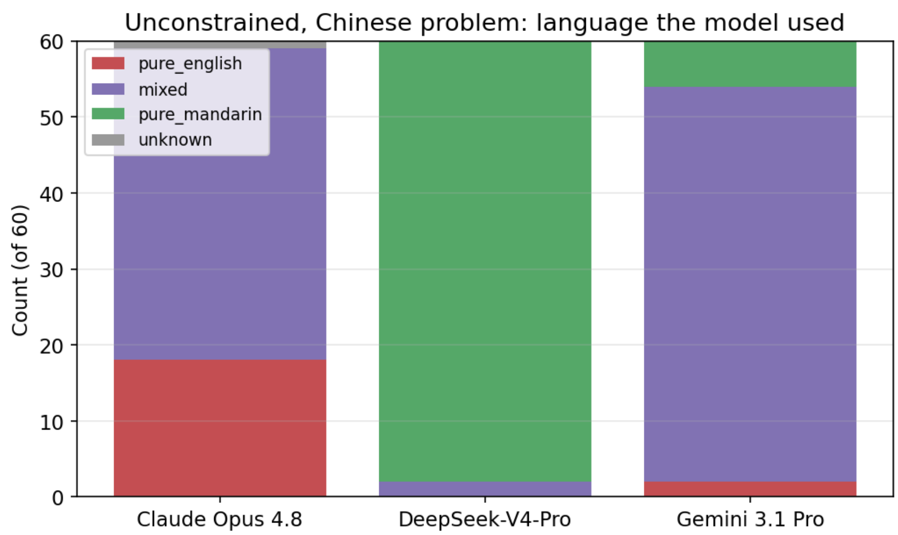

Figure 2: Language used on unconstrained Chinese-original problems ($n = 60$). DeepSeek's raw trace stays largely in Mandarin; the summarized Claude and Gemini outputs skew English or mixed.

4 Discussion

Ultimately, these patterns demonstrate that constraining a model's language of reasoning affects its accuracy minimally, if at all—often, the constraints would be disregarded or not obeyed by models altogether, leading to a lessened impact on performance.

On the point of obedience, the Chinese model DeepSeek was (unsurprisingly) the most likely to agree to reasoning in Chinese when prompted to do so, and the Western-developed Claude the least so. The frequent refusal of models to reason in Chinese is ostensibly the result of a majority of English writing and information in the pool of training data.

References

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